

HOME AUTOMATION SYSTEM For Up To Four Devices

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This project is designed to control up to four home electrical devices using Node-RED, Raspberry Pi and local Wi-Fi network. Node-RED is a flow-based development tool for visual programming.

Node-RED

Node-RED is a graphical user interface (GUI)-based programming platform. It is a powerful platform to make Internet of Things (IoT) applications

just by connecting its predefined code blocks like input, output, social media, Raspberry Pi GPIO and so on. You just need to connect predefined code blocks, called nodes, to perform a task. Connected nodes are collectively called flow.

Node-RED was developed as an open source project at IBM in 2013 to connect their hardware with Web-based applications, email servers, social media, etc. It has now become a popular IoT programming tool. When combined with Raspberry Pi, Node-RED becomes even more efficient for interacting with the physical world.

Preparing Raspberry Pi for Node-RED

The latest Raspbian operating system (OS) for Raspberry Pi comes with Node-RED application by default. If you want to install the latest version of Node-RED, run the following commands on the terminal:

```
sudo apt-get update
sudo apt-get upgrade
```

This will install the latest Node-RED on Raspberry Pi. If you are facing problems, install it manually by entering the following script on the terminal:

```
bash <(curl -sL https://raw.githubusercontent.com/node-red/raspbian-deb-package/master/resources/update-nodejs-and-nodered)
```

Note that, after using this script to upgrade Node-RED, you will not be able to upgrade it again using apt-get upgrade command.

To open Node-RED from the desktop of Raspberry Pi, click on Raspberry Pi icon and then from the drop-down menu select Programming→Node-RED. Alternatively, you can type the following command on the terminal:

```
Node-RED console
File Edit Tabs Help
sudo npm i -g npm@2.x

Started Node-RED graphical event wiring tool..
Welcome to Node-RED
=====
29 Aug 15:04:51 - [info] Node-RED version: v0.14.6
29 Aug 15:04:51 - [info] Node.js version: v0.10.29
29 Aug 15:04:51 - [info] Linux 4.4.13-v7+ arm LE
29 Aug 15:04:51 - [info] Loading palette nodes
pi : TTY=unknown ; PWD=/ ; USER=root ; COMMAND=/usr/bin/python -u /usr/lib/node_modules/node-red/nodes/core/hardware/nrgpio.py info
pam_unix(sudo:session): session opened for user root by (uid=0)
pam_unix(sudo:session): session closed for user root
29 Aug 15:05:00 - [info] UI started at /ui
29 Aug 15:05:06 - [info] Settings file : /home/pi/.node-red/settings.js
29 Aug 15:05:06 - [info] User directory : /home/pi/.node-red
29 Aug 15:05:06 - [info] Flows file : /home/pi/.node-red/flows_raspberrypi.json
29 Aug 15:05:06 - [info] Server now running at http://127.0.0.1:1880/
29 Aug 15:05:06 - [info] Starting flows
29 Aug 15:05:06 - [info] [nodebot:Raspberry pi 2] error loading io class
pi : TTY=unknown ; PWD=/ ; USER=root ; COMMAND=/usr/bin/python -u /usr/lib/node_modules/node-red/nodes/core/hardware/nrgpio.py out 38
pam_unix(sudo:session): session opened for user root by (uid=0)
```

Fig. 1: Node-RED console

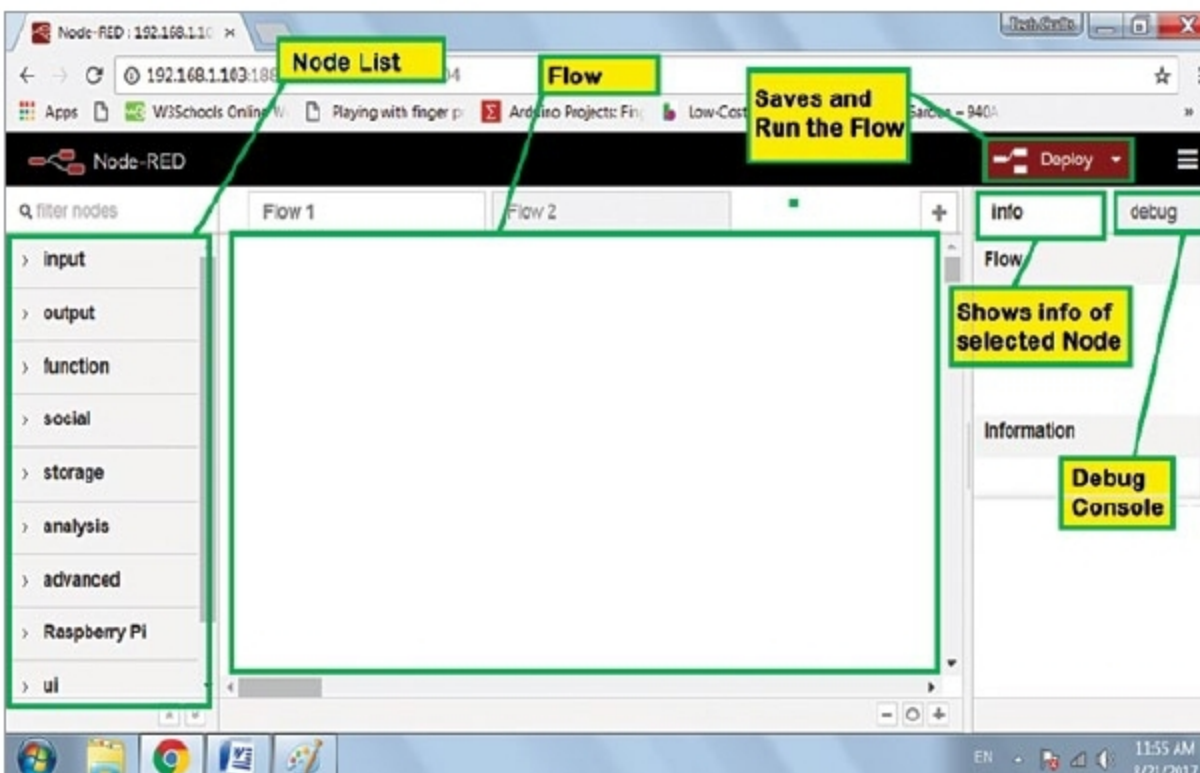


Fig. 2: Homepage of Node-RED